

Q Given an array.
Rotate the array from last to first K times

A: 3 -2 1 4 6 9 8

$K=1$ 8 3 -2 1 4 6 9

$K=2$ 9 8 3 -2 1 4 6

$K=3$ 6 9 8 3 -2 1 4

I) BF

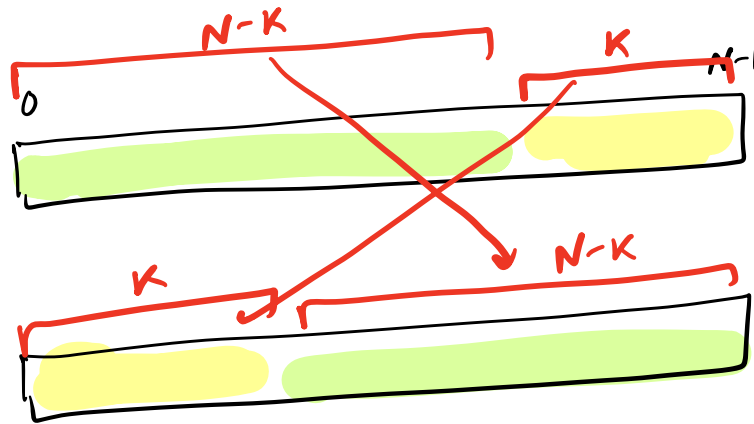
$N \leq 10^5$
 $K \leq 10^9$

1 rot $\rightarrow O(N)$
 $K - \rightarrow O(NK)$

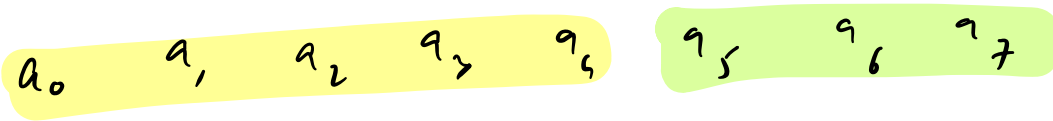
10^{14} ops

II

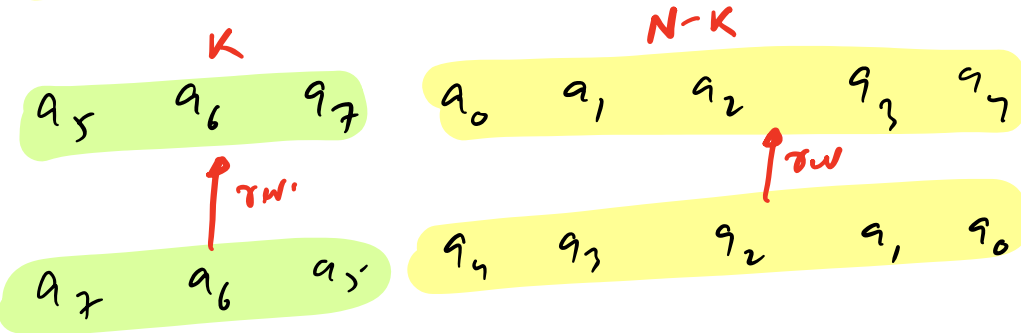
After K
rot



A:



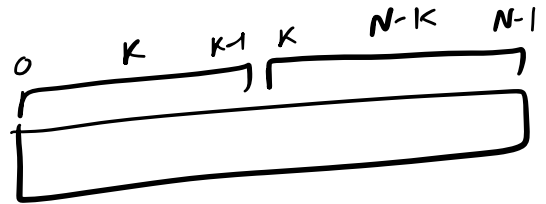
K=3



STEPS:

1. Reverse the whole Array ✓
2. Reverse first K elements ✓
3. Reverse last N-K — ✓

✓ reverse(A, 0, N-1);
 ✓ ——— reverse(A, 0, K-1);
 ✓ ——— reverse(A, K, N-1);



SC = O(1)

TC = O(N)

$K \geq N$?

$\% 3$

A

$N=3$

$K=0$, $3, 6, 9$ - $[1, 2, 3]$

$K=1$, $4, 7, 10$ - $[3, 1, 2]$

$K=2$, $5, 8, 11$ - $[2, 3, 1]$

$K=10$

$$K = K \% N$$

$10 \% 3$

$K=1$

Φ

$$\begin{aligned} & \text{for } (i=1; i \leq n; i++) \{ \\ & \quad \text{for } (j=1; j \leq 3^i; j++) \{ \\ & \quad \quad \quad \equiv \\ & \quad \quad \quad \equiv \end{aligned}$$

}

}

$\begin{aligned} & O(2^N) \\ & O(3^N) \\ & O(N^2) \\ & O(N \log N) \end{aligned}$

i	j [1 - 3 ⁱ]	# it
1	[1 - 3 ¹]	+ 3 ¹
2	[1 - 3 ²]	+ 3 ²
3	[1 - 3 ³]	+ 3 ³
⋮	⋮	⋮
N	[1 - 3 ^N]	+ 3 ^N

$$3^1 + 3^2 + 3^3 + \dots + 3^N$$

$\xrightarrow{\times 3}$ $\xrightarrow{\times 3}$

$$\frac{a[r^t - 1]}{r - 1}$$

GP

$$a: 3$$

$$r = 3$$

$$t = N$$

$$\frac{3[3^N - 1]}{3 - 1}$$

$$\frac{3}{2}[3^N - 1]$$

$$\cancel{\frac{3}{2} \cdot 3^N} - \cancel{\frac{3}{2}}$$

$TC = O(3^N)$

Q

$$A: [\overset{0}{1}, \overset{1}{-1}, \overset{2}{0}]$$

$$B: [1, 0, 1]$$

$$PS: [1, 1, 2]$$

non-negative

Q: $[L, R]$

$$[0, 2] \rightarrow 2$$

$$[1, 2] \rightarrow 1$$

$sum[L, R] = ps[R] - ps[L-1]$

$\swarrow$
 $L=0: 0$

non-neg $\rightarrow +1$
neg $\rightarrow 0$

STEPS:

1. Create an array $B[] \rightarrow O(N)$
s.t. $B[i] = 1 \rightarrow A[i]$ is non-neg.
 $B[i] = 0 \rightarrow$ negative
2. Build the $PS[]$ for $B[] \rightarrow O(N)$
3. Answer queries using the $PS[] \rightarrow O(Q)$

~~$TC = O(N+Q)$~~

~~$SC = O(N)$~~

$O(1)$